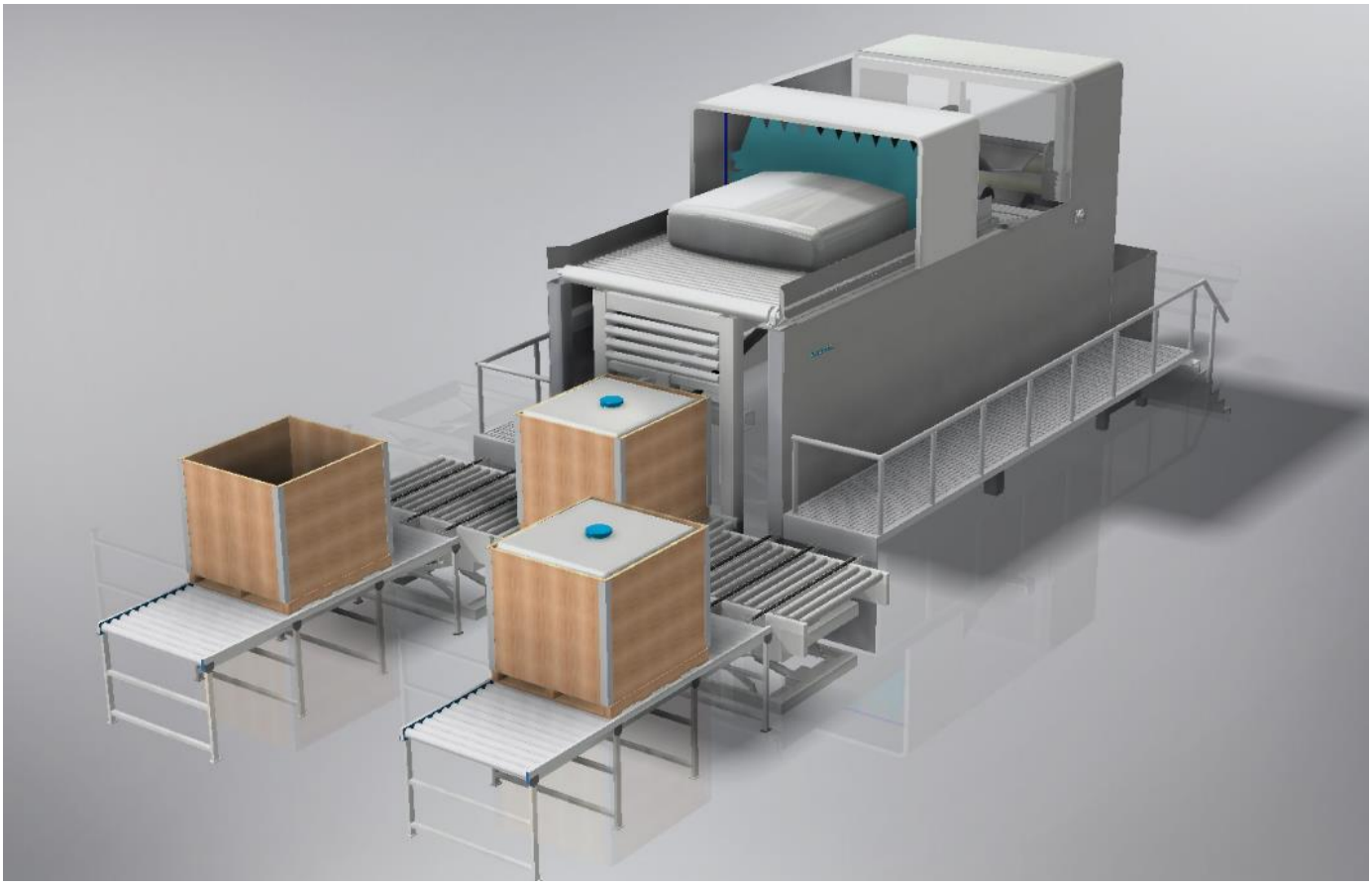
anantausa

UNLOADING MACHINE
Tote Bins Dumper



Contents Table

1 Objective 3

2 Machine description 3

2.1 Working Method 3

2.2 Constructive Structure 4

2.2.1 Common features 4
2.2.2 Roller Conveyors 4
2.2.3 Dumping Section 5
2.2.4 Cutting and Emptying Section 6
2.2.5 CIP System 7
2.2.6 Control and Auxiliary Systems 7

3 Other Equipment 8

3.1 Additional Roller Conveyors 8

3.2 Remote control module 8

4 Conclusions 9

1 Objective

The purpose of this brochure is to provide a description of the components and functionalities of the Unloading Machine - Tote Bins Dumper carried out by ANANTA USA.

2 Machine description

2.1 Working Method

This machine is capable of damping one tote bin per each movement. In order to achieve this, the bins with the aseptic bags will be placed in the transportation tables to be transported by roller conveyors. At this point, the bins must be opened and any obstacle that could be an obstacle for the bag extraction must be removed.

Once the tote bin is in the dumping section, a cylinder elevates the bin and turns it over; the bin is fixed to the structure by blocking systems located in the upper edge of the container. These systems avoid the tote bin from moving out of the structure during the turn over phase that extracts the aseptic bag that contains the raw materials.

When the bag is extracted and located in the cutting and emptying section, the dumping structure goes down with the empty tote bin and it is removed from the structure by the roller conveyors.

At the same time, the aseptic bag goes through the cutting and emptying section. Firstly, it is washed by pressurized water with the use of diffusers located all around the aseptic bag. To remove this water from the surface, the bag passes through a plastic curtain that swipes this water. Later in the process, the bag passes under a cylinder that presses the raw material to the bottom of the bag.

After this process we have a clean and ready to be cut aseptic bag. This cut is made by mechanical rotary blades that are rolling in the same direction as the advancement of the bag to avoid any trapping.

As the aseptic bag moves over the blades, it goes through a roller press located above a hopper, this press forces the inner raw material out of the bag to the hopper, where it is collected and stored until it is needed in the production process.

The empty bags fall to a removable container for a later disposal.

When the raw material is sent to the process line, it will be mixed with water in the proportion adequate for the ´recipe´ in production. This will be carried out by a fully automated system.

In order to wash of the machine, a specialized system is provided. This system cleans the parts of the machine which are in contact with the product.

2.2 Constructive Structure

2.2.1 Common features

All the phases are carried out in a sequence controlled by the main PLC of the machine.

All components are made of stainless steel AISI 304, except:

- Cutting blades and product hopper in AISI 316 with mirror finish.
- Some standard trading components.

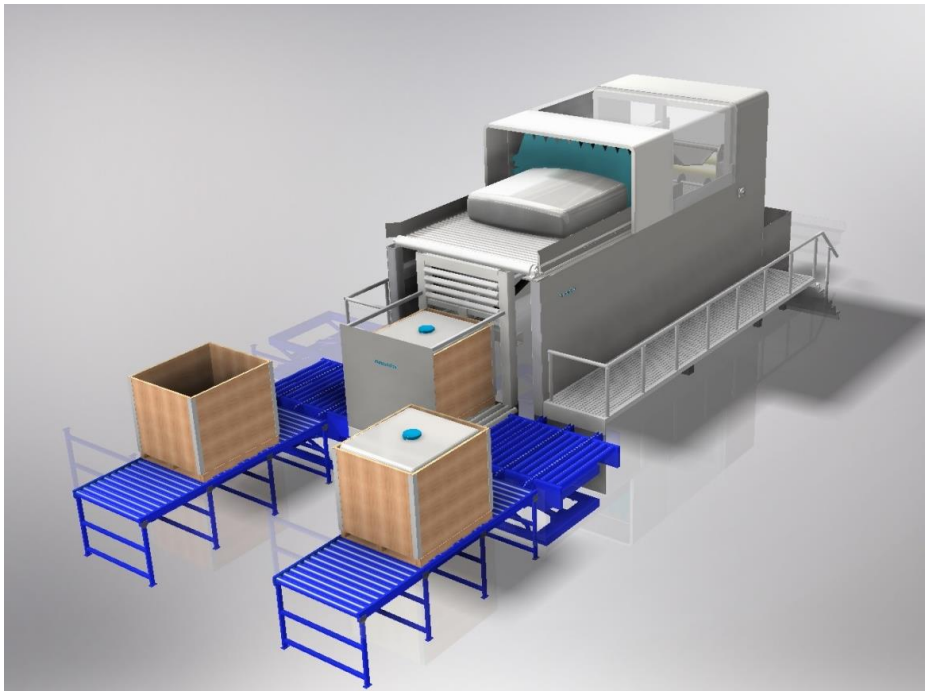
2.2.2 Roller Conveyors

This equipment has been designed to provide a buffer of filled tote bins to the machine and another buffer of empty tote bins.

This roller conveyor is made in stainless steel AISI304L. It has a "*U*" layout, in order to have a buffer of 3 filled tote bins and another buffer of 3 empty tote bins.

Its dimensions are 118" long and 47" wide in the buffer zones and 155" long and 47" wide in the dumping section zone.

These conveyors have a forklift protection system in order to avoid damages made by collisions and are driven by electric engines, controlled by the main PLC of the machine.



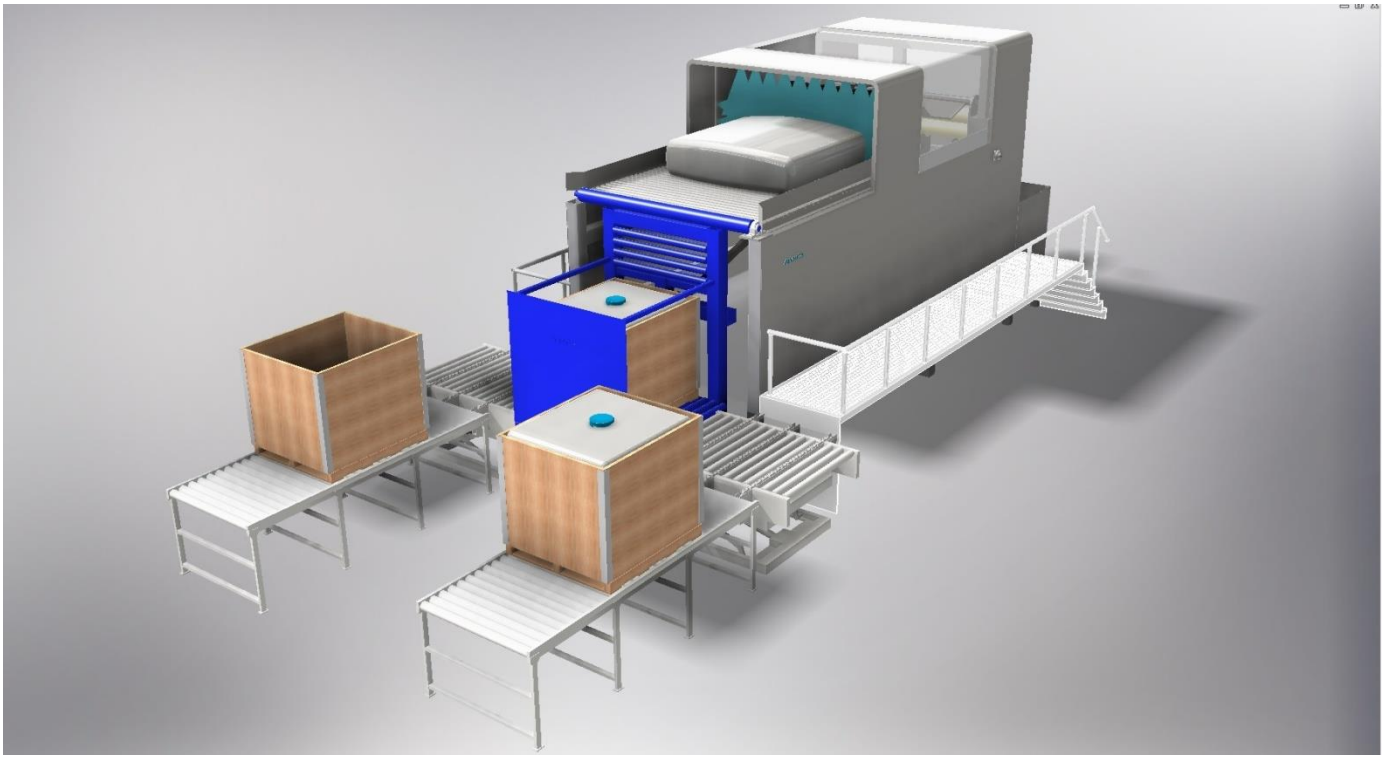
2.2.3 Dumping Section

This section has the capacity to unload up to 20 bins per hour.

It has a security system that prevent the tote bin from falling on to the transportation when it is turned over and lateral barriers to keep the tote bin in position when the dumping movement is being executed.

The movement of the bins is made by a cylinder located under the main structure and has power to lift and lower up to 4000lbs.

The transport of the tote bins in and out of the structure, is made by a motorized roller conveyor.



2.2.4 Cutting and Emptying Section

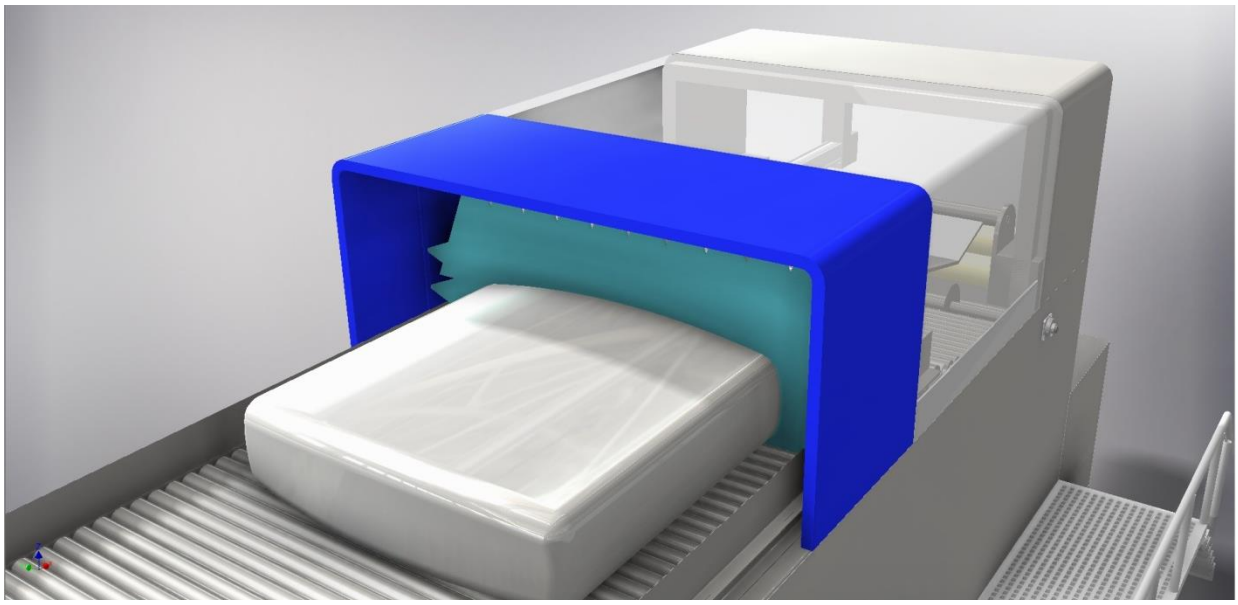
These conveyors belt is 906” wide. This section is composed of three parts:

2.2.4.1 Washing

The washing is made by 14 diffusers with pressurized water at 22 psi that removes any dirt that the aseptic bag could have. These diffusers are located all around the bag.

The water isolation from the rest of the machine and its removal is done by a plastic curtain.

The pressing cylinder located after this curtain is placed there to move the raw material to bottom of the bag and is made in stainless steel AISI 304L.



2.2.4.2 Cutting

The sharp blades used to cut the aseptic bag are made in stainless steel AISI 316L, and roll in the same direction as the advance of the bag to avoid trapping it.

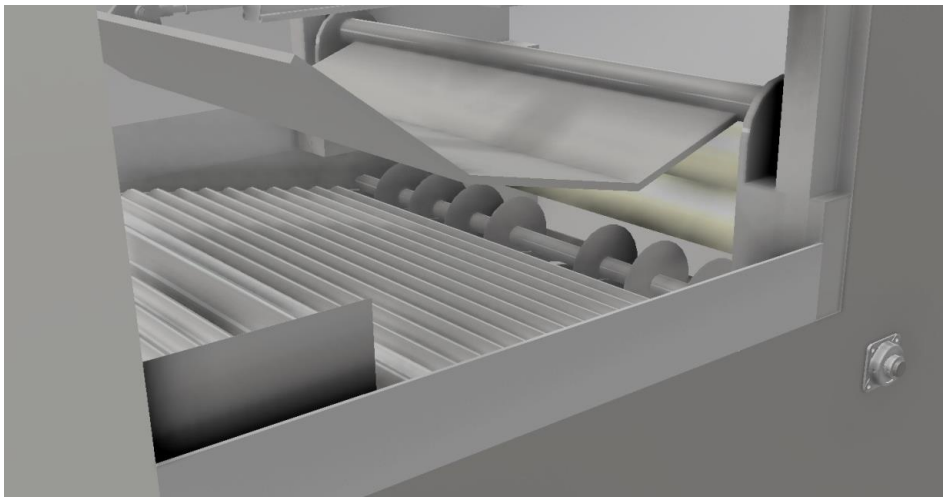
They also have a plain edge (no saw-tooth), this makes a fine cut, avoiding the possibility of throwing bag portions to the product hopper.

The position of the blades in the machine enables an easy maintenance.

2.2.4.3 Emptying

The roller press that extracts the raw material minimizes the raw material lost.

The product hopper is located over a pump (provided by the client) to send the raw material to the production process line.



2.2.5 CIP System

The CIP system is composed of pipes that flow pressurized water to all parts of the machine in contact with the product.

The system is connected to the main water supply of the machine. It cleans the machine with pressurized water which removes the dirt of the machine components. The waste water is collected in the hopper and drained from here to disposal.

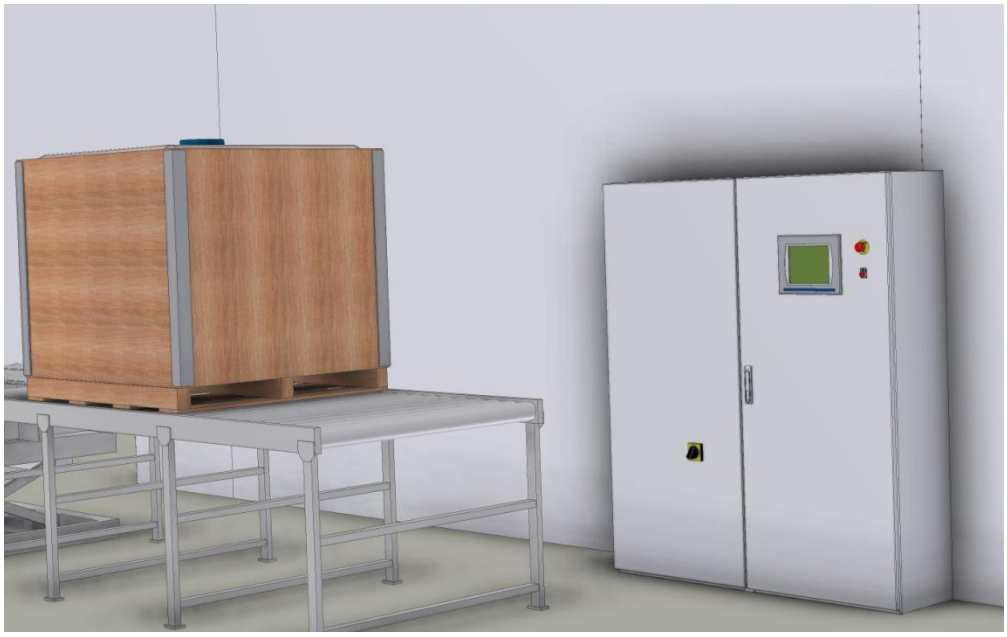
2.2.6 Control and Auxiliary Systems

The machine is controlled by a main PLC (Programmable Logic Controller), which is connected to several sensors in the machine that enable this controller to know exactly what is happening in the machine. With this information and based on the programmed software with the PLC, the controller will execute different actions in response of the state of the sequence active.

As this controller has industrial communication capabilities, it can be connected to the main control system of the factory in order to get a telematic control of the unit or make a historic data base. The system also has a flat touch panel to make the Human-Machine interface.

All the electric components have a selective overcurrent protection.

The machine posses a safety system which is categorized at SIL3 by the EN/ISO 62061.



3 Other Equipment

3.1 Additional Roller Conveyors

Additional roller conveyor can be added to increase the amount of these buffers to 5 tote bins, limit based on the dimensions of the room where the machine would be placed.

3.2 Remote control module

As it was said in the point “Control and Auxiliary Systems”, this machine can be remotely controlled or supervised.

To achieve this, an electronic device can be added to the machine. This remote control module enables the PLC of the machine to communicate with SCADA systems or historic data servers.

4 Conclusions

The Unloading Machine **increases the operative performance** of the plant and optimizes the unloading process. It **reduces product loses** to virtually cero and significantly **reduces the necessary manpower**.

With the Unloading Machine only one person is necessary per shift. The employee needs to upload the reception area with as many containers as there is room available and takes the lids off. Once this is done, the person can move on to other activities until another load is necessary. Current methodologies of unloading such as suction takes 2 to 4 people per shift and the personnel needs to be dedicated full-time to the unloading process.

Current Estimated Losses of product due to current unloading mechanisms range from 3% to 20%. The proposed process virtually leaves no product behind (0.5% is used for comparative purposes).

Given the virtually cero product losses and the reduction in personnel the unloading machine has **very low payback periods**.

The Tote Bin Dumper:

- Optimizes the unloading of tote bins
- Has an adjustable capacity of up to 20 tote Bins per Hour
- Includes aseptic bags washing system
- Will reduce product loses to virtually cero
- Is a fully automated system